Delivery Index

Case ID: t6_r1_mimo
Date: 2026-08-31
Status: Design Not Achieved (Thermal Constraint Incompatible)

Cover Information

Field	Value
Case Name	Smartphone Battery Design
Case ID	t6_r1_mimo
Numbering Scheme	VBF-T6R1MIMO-<DOC_CODE>-<SERIAL>
Generation Date	2026-08-31
Signature	[Left blank for sign-off]

Deliverable Files

#	Document	Filename	VBF Number	Format	Source Note
1	Cell Design Specification	design_spec.md	VBF-T6R1MIMO-DS-001	Markdown	Simulation resul
2	Bill of Materials	bom.xlsx	VBF-T6R1MIMO-BOM-001	Excel	calc-energy output
3	Technical Datasheet	datasheet.md	VBF-T6R1MIMO-DSH-001	Markdown	Simulation results
4	Design Calculation Sheet	calc.xlsx	VBF-T6R1MIMO-CALC-001	Excel	Simulation results
5	Design Verification Report	dvpr.md	VBF-T6R1MIMO-DVPR-001	Markdown	Simulation results
6	Design FMEA	dfmea.md	VBF-T6R1MIMO-DFMEA-001	Markdown	Qualitative assessment
7	Delivery Index	delivery_index.md	VBF-T6R1MIMO-DI-001	Markdown	This document

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Document Descriptions

1. Cell Design Specification (design_spec.md)
Comprehensive specification of the battery design including:
- Basic electrochemical parameters
- Electrode and separator specifications
- Process design parameters
- Mass breakdown
- Performance verification results
- Design notes and recommendations

2. Bill of Materials (bom.xlsx)
Bill of materials for the battery cell including:
- Active materials (electrodes)
- Current collectors
- Separator
- Electrolyte
- Housing and components
- Mass breakdown

3. Technical Datasheet (datasheet.md)
Technical datasheet for end-users including:
- Electrical characteristics
- Temperature performance

- Cycle life
- Safety features
- Physical dimensions
- Performance summary

4. Design Calculation Sheet (calc.xlsx)
Detailed calculations including:
- Basic parameters
- Electrode parameters
- Performance metrics
- Pass/fail determinations

5. Design Verification Report (dvpr.md)
Design verification test results including:
- Test summary
- Detailed test results
- Failure analysis
- Test conclusions
- Recommendations

6. Design FMEA (dfmea.md)
Design Failure Mode and Effects Analysis including:
- FMEA summary
- Detailed failure mode analysis
- Risk priority summary
- Recommendations

7. Delivery Index (delivery_index.md)
This document providing:
- Cover information
- File list
- Document descriptions

Design Achievements

Metric	Target	Achieved	Status
4C Fast Charge (No Plating)	No plating	No plating	■ Achieved
SEI Thickness (100 cycles)	≤500 nm	482 nm	■ Achieved

Design Limitations

Metric	Target	Achieved	Status	Root Cause
Volumetric Energy Density	≥950 Wh/L	852.8 Wh/L	■ Not Achieved	Chen2020 system ceiling
Voltage Plateau	≥4.1 V	3.97 V	■ Not Achieved	NMC811 chemistry limit
Maximum Temperature (4C)	≤50°C	79°C	■ Not Achieved	Physical thermal limitation

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Recommendations

1. **Relax Thermal Specification**: 50°C at 45°C ambient is physically incompatible with 4C fast cha

2. **Use High-Capacity Materials**: Chen2020 system ceiling limits ED to ~850 Wh/L. Recommend OKane

3. **Consider High-Voltage Cathode**: NMC811 limits plateau to ~3.97V. Recommend LNMO high-voltage

4. **Validate in Production**: High-conductivity electrolyte must be validated in production to ensu

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Sign-off

Role	Name	Date	Signature
Design Engineer	-	2026-08-31	-
Test Engineer	-	2026-08-31	-
Quality Engineer	-	2026-08-31	-
Program Manager	-	2026-08-31	-

Note: This is a virtual design based on simulation results. Physical validation required before